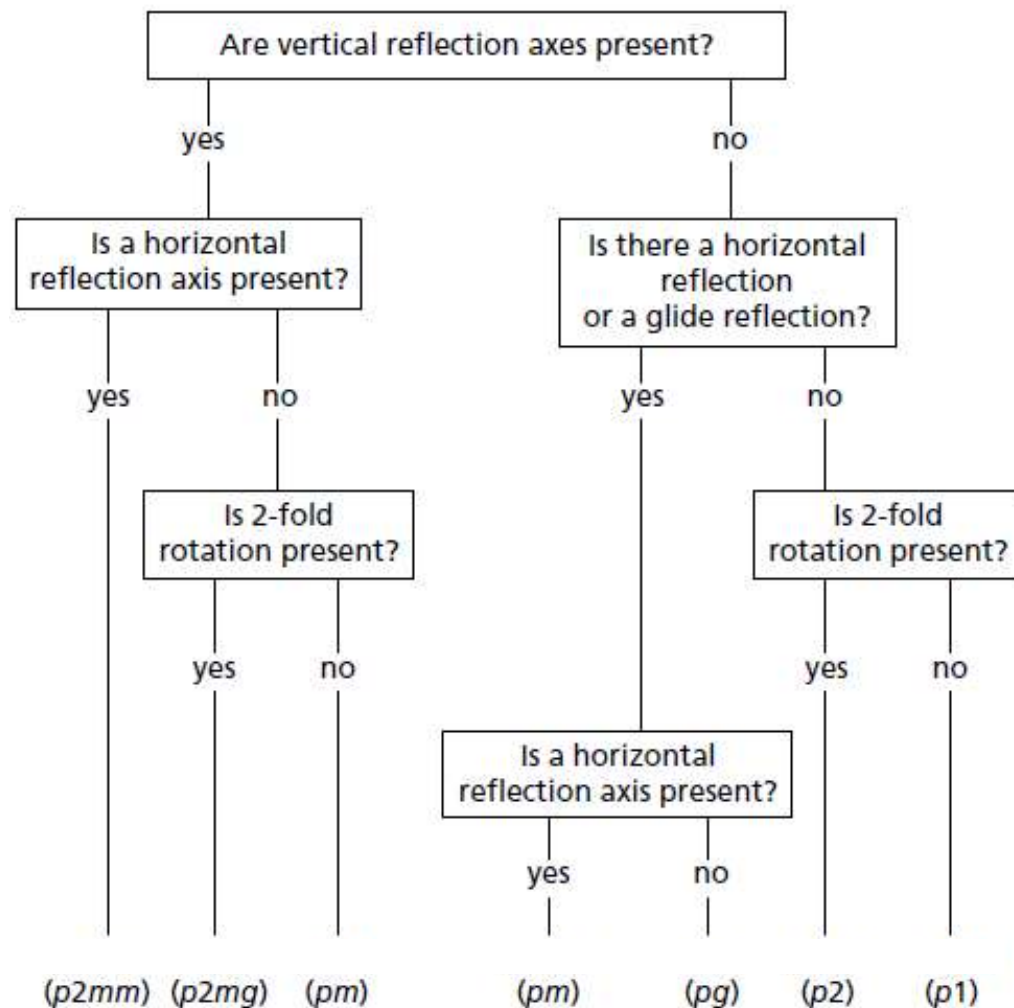


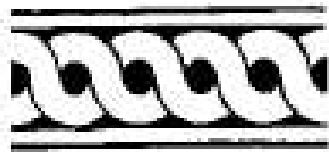
1D Groups - Flowchart



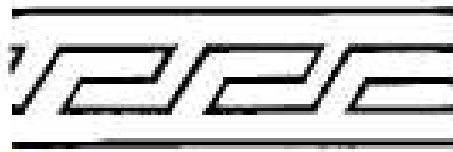
Six unique Line Groups:
p1, p2, pm, pg, p2mg, p2mm

In-Class Exercise 1

Find the unit cell, lattice, symmetry elements, and line group.



(a)



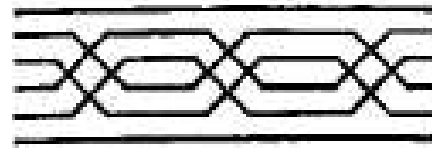
(b)



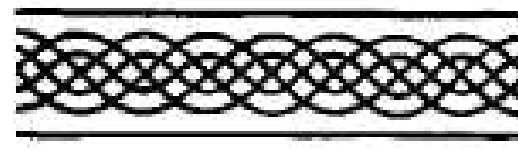
(c)



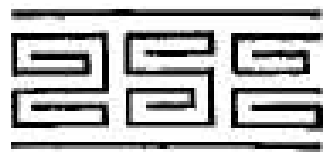
(d)



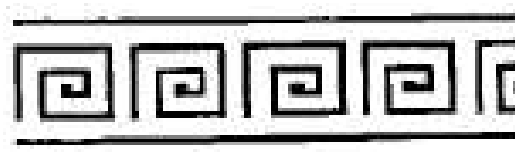
(e)



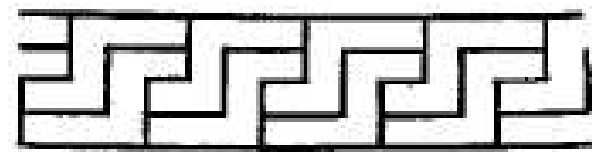
(f)



(g)



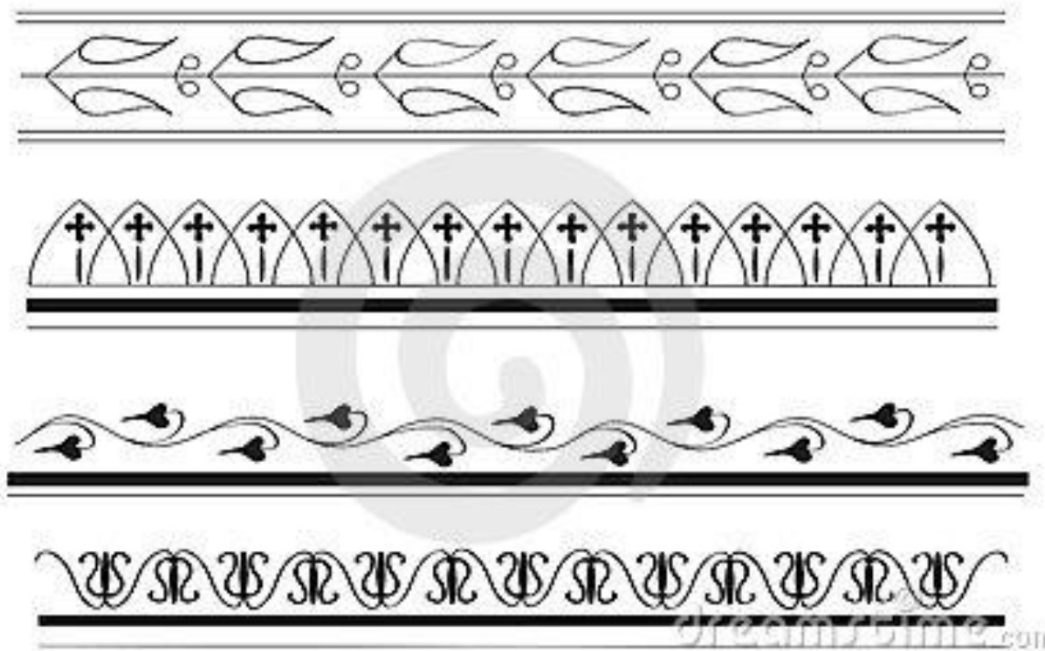
(h)



(i)

In-Class Exercise 2

Identify the unit cell, as many symmetry elements as you can and the point group of the following structures.

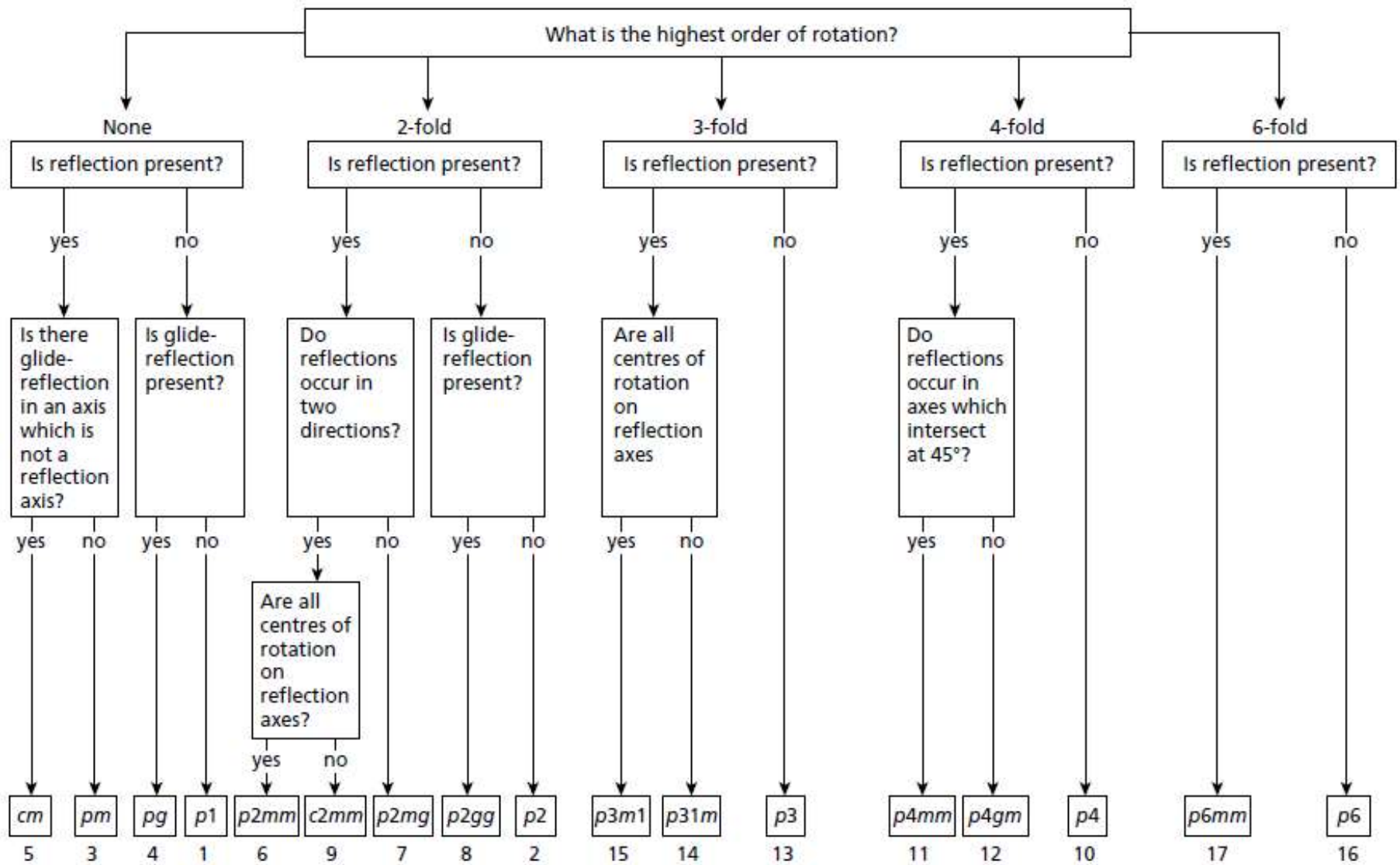


In-Class Exercise 3

Identify as many symmetry elements as you can and find the point group of the following images.

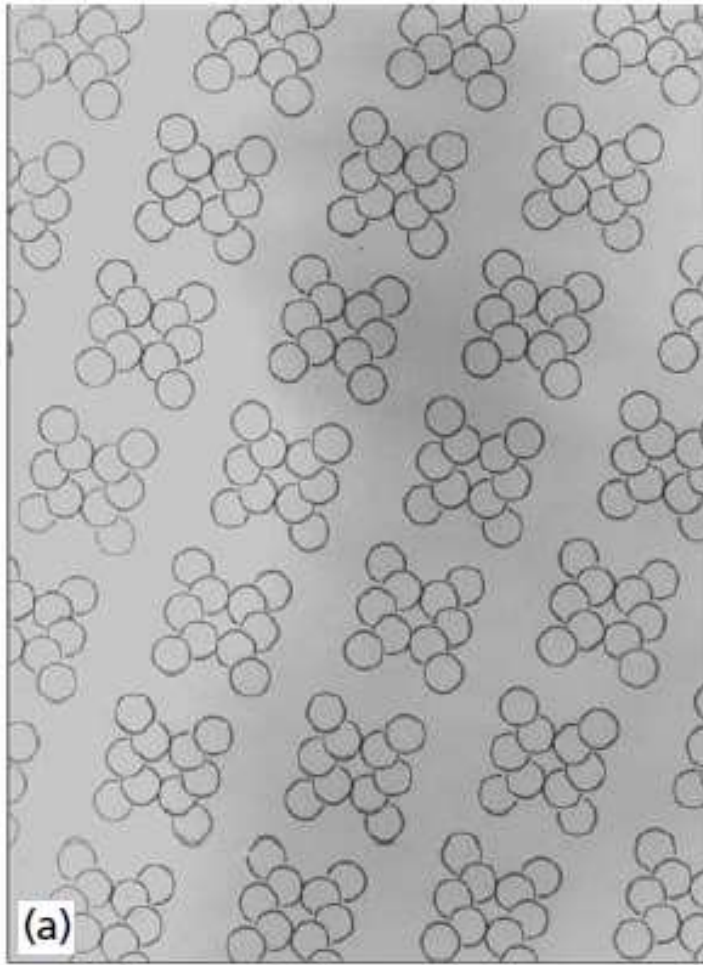


Plane Groups Flowchart



In-Class Exercise 4

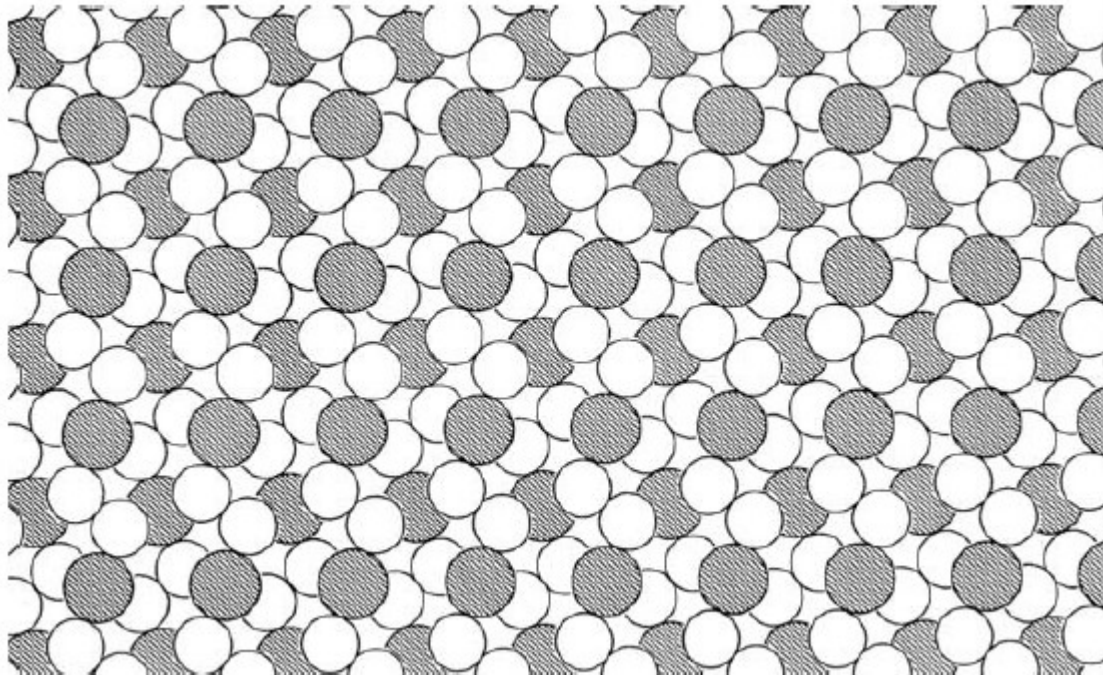
Recognizing unit cells & plane groups



- (1) Identify basis
- (2) Identify lattice (translation)
- (3) Identify rotation
- (4) Identify mirror

In-Class Exercise 5

Recognizing unit cells & plane groups
FeS₂, Marcasite



In-Class Exercise 6

Identify the unit cell, as many symmetry elements as you can and the point group of the following structures.

